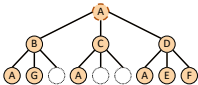


Step 1: Graph Structure -> Node Sequence

Neighborhood Detail Template

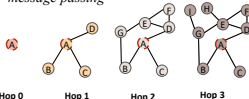


Node Sequence:
A B C D A G [pad] A [pad] [pad] A E F

or

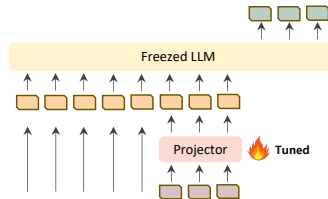
Hop-Field Overview Template

message passing



Node Sequence:
[Hop 0] [Hop 1] [Hop 2] [Hop 3]

Step 2: Node Sequence -> Token Embedding Sequence



Prompt Design (Base Model: Vicuna-v1.5)

SYSTEM MESSAGE: A chat between a curious user and an artificial intelligence assistant. The assistant gives helpful, detailed, and polite answers to the user's questions.

USER: Given you a node: <node sequence>...Please tell me...

ASSISTANT: <Answer>

Training Prompt: [SYSTEM MESSAGE] USER:[USER]

ASSISTANT: [ASSISTANT] </STOP>

Inference Prompt: [SYSTEM MESSAGE] USER:[USER]

ASSISTANT:

Training Task

Node Classification

Given a node-centered graph: <node sequence>, we need to classify the center node into 40 classes: cs.NA(Numerical Analysis), cs.MM(Multimedia), ..., please tell me which class the center node belongs to?

cs.NA(Numerical Analysis)

Link Prediction

Given two node-centered subgraphs: <node sequence> and <node sequence>... Please tell me whether two center nodes in the subgraphs should connect to each other. Answer yes or no.

No

Node Description

Please describe the center node : <node sequence>.

The center node represents a paper in cs.AI(Artificial Intelligence), it's about simultaneous merging multiple grid maps using the robust motion averaging.

Figure 1. Illustration of LLaGA framework and its prompt design paradigm.

central node and its immediate surroundings. The second, the **Hop-Field Overview Template**, offers a summarized view of a node's neighborhood, extendable to larger fields.

Neighborhood Detail Template is designed to elaborate on the detailed information of a node and its surrounding

to a relative structural position within the original graph.

Post conversion of the center node and its structural information into a node sequence, we shift to mapping them into the node embedding space. In the context of text-attributed graphs, we can utilize various off-the-shelf text encoding